

# Software Requirements and Dependencies

## 1. Runtime Environment:

This project is developed and optimized for Google Colab using a standard Python 3.10+ runtime. Using this cloud-based platform ensures consistency in the execution environment and provides immediate access to necessary hardware acceleration without local configuration.

## 2. Hardware Recommendations:

The provided Colab notebook is highly flexible and can be executed using either of the following hardware configurations:

- **Option A: GPU (Highly Recommended):** A NVIDIA T4 GPU (available within Colab via Runtime -> Change runtime type) is strongly recommended. It significantly accelerates the dense matrix multiplications required for SBERT embedding generation and the iterative training of the Multi-Layer Perceptron (MLP) baseline.
- **Option B: CPU (Supported but Slower):** The notebook is fully compatible with standard CPU runtimes. However, execution time will increase substantially, particularly during the neural network training phases.

## 3. Core Libraries and Dependencies:

The majority of the required libraries are natively pre-installed in the Google Colab environment. The core packages utilized include:

- **pandas, numpy:** For data manipulation and results export.
- **scikit-learn:** Extensively used throughout the pipeline for
  - feature engineering: TfidfVectorizer
  - model tuning: GridSearchCV
  - Classifiers: GaussianNB, MLP, SVM
  - evaluation metrics: accuracy\_score, precision\_score, recall\_score, f1\_score, roc\_curve, auc
- **nltk:** For text preprocessing and stopwords filtering.
- **scipy:** For statistical significance testing (Wilcoxon signed-rank test).
- **matplotlib, seaborn:** For generating the final performance and significance visualizations.

## 4. External Libraries:

The following library is required for semantic feature extraction and is not pre-installed:

- **sentence-transformers:** Used to load the 'all-MiniLM-L6-v2' model for SBERT embeddings.

## 5. Network Requirement:

An active internet connection is strictly required during runtime. The script dynamically fetches the bug report datasets via raw GitHub URLs and downloads the pre-trained SBERT model

weights from the Hugging Face hub.

**Installation Note:**

No manual local environment configuration is required. The first cell of the provided Jupyter Notebook (.ipynb) automatically handles the download of NLTK stopwords and the installation of sentence-transformers via a !pip install command upon execution.